(c) The spacecraft now splits apart into a carrier and a payload as shown in Fig. 2.3.

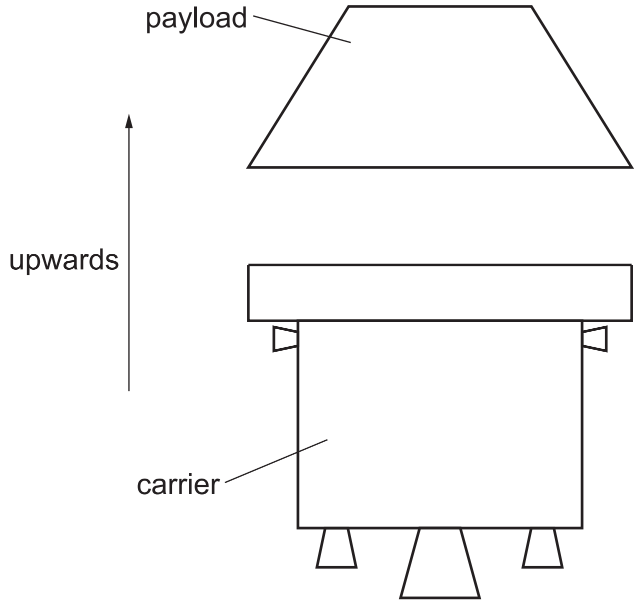

Fig. 2.3

During the split, an average force of 5500 N acts on the payload for a time of 0.36 s. The velocity of the payload increases by $8.5\,\mathrm{m\,s^{-1}}$ in the upwards direction.

The combined mass of the carrier and payload is 2.5×10^3 kg.

(i) State the principle of conservation of momentum.

....

....

.... [2]

(ii) Show that the mass of the payload is 230 kg.

[2]